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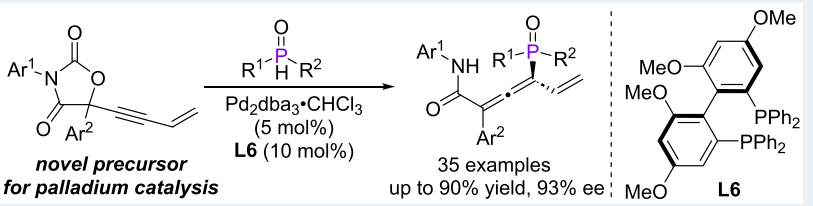
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Pd-Catalyzed Asymmetric Allenylation of Secondary Phosphine Oxides with Enyne-Type Propargylic Carbamates for the Construction of Chiral Allenyl Phosphine Oxides

Yujie Dong, Nianci Zhang, Fazhou Yang, Jinbao Wang, Bo Wang, Jun Liu, Bing Zheng,* Cheng Zhang,* Leijie Zhou,* and Hongchao Guo*

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ABSTRACT: Chiral allenyl phosphine oxides represent a significant kind of synthetic target with promising potential for various applications. However, explorations toward their asymmetric synthesis remain scarce, primarily due to the competitive coordination of phosphine oxides with metal catalysts. Herein, we report a Pd-catalyzed enantioselective allenylation of secondary phosphine oxide for the efficient construction of a variety of chiral allenyl phosphine oxides. The success of this reaction arises from the enhanced (bidentate form) coordination of a newly designed, fungicide-inspired enyne-type propargylic carbamate with a chiral Pd complex. This protocol features high yields and enantioselectivities, scalability, and versatile derivatization. Mechanistic studies confirmed the exclusion of a kinetic resolution pathway.

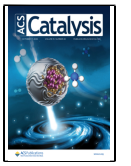
KEYWORDS: *asymmetric catalysis, palladium catalysis, phosphine oxide, chiral allene, axial chirality*

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INTRODUCTION

Chiral allenes are privileged motifs in natural products, bioactive molecules, and functional materials.[1] Consequently, recent decades have witnessed significant advances in the development of catalytic methods toward their asymmetric synthesis.[2] As a result, a wide variety of chiral allenes bearing diverse functional groups could be obtained conveniently through asymmetric catalysis approaches.[3] On the other hand, phosphine oxides that are incorporated within various chiral scaffolds are widely used as organocatalysts and ligands for metal catalysis (Figure 1a). Related synthesis has attracted more and more attention.[4,5] Despite elegant progresses that have been made in both areas, a method to access chiral allenyl phosphine oxides remains unexplored until very recently, Ma, Zhang, Qian, and Zhang jointly published the first Pd-catalyzed enantioselective allenylation of secondary phosphine oxides (SPOs) (during which we are also preparing this manuscript).[6] Previously, Ma, Zhang, and Qian pioneered the asymmetric allenylation of *H*-phosphonates (2022), representing the sole, highly valuable example of a catalytic approach to axially chiral allenyl phosphonates.[7] This breakthrough suggested that SPOs might also serve as viable substrates for the concise construction of chiral allenyl phosphine oxides via transition-metal-catalyzed asymmetric allenylation reaction. However, SPOs undergo facile tautomerization to trivalent R^1R^2P-OH species more readily than *H*-phosphonates, leading to competitive metal coordination that suppresses the desired allenylation pathway (Figure 1b).[8] To address this, several elegant strategies were developed to avoid the involvement of trivalent R^1R^2P-OH species. Guo presented a Ni-catalyzed O-propargylation/Horner-Mark cascade to construct chiral allenyl phosphine oxides directly.[9] Meanwhile, Zhang developed a Ni-catalyzed asymmetric S_N2' functionalization of phosphines, with subsequent oxidation delivering chiral allenyl phosphine oxides.[10] He reported an asymmetric synergistic Pd/Co-catalyzed hydrophosphinylation of enynes using a newly designed masked phosphinylating reagent, enabling efficient access to novel allenylic phosphine oxides with *P*-stereocenter.[11] Despite these advances, methods for direct allenyl $C(sp^2)-P$ bond construction remain highly sought-after yet scarce, presenting a persistent synthetic challenge.

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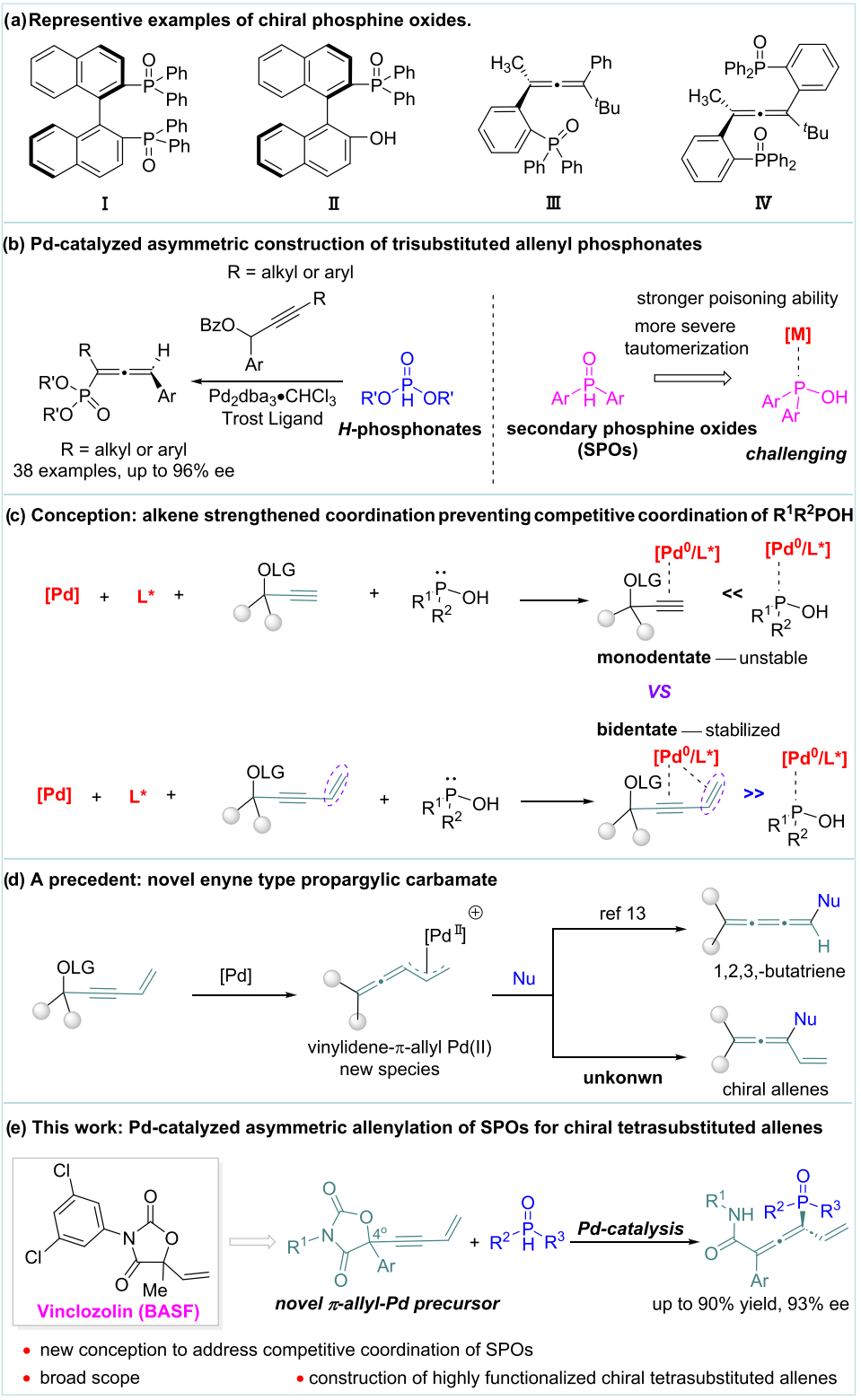
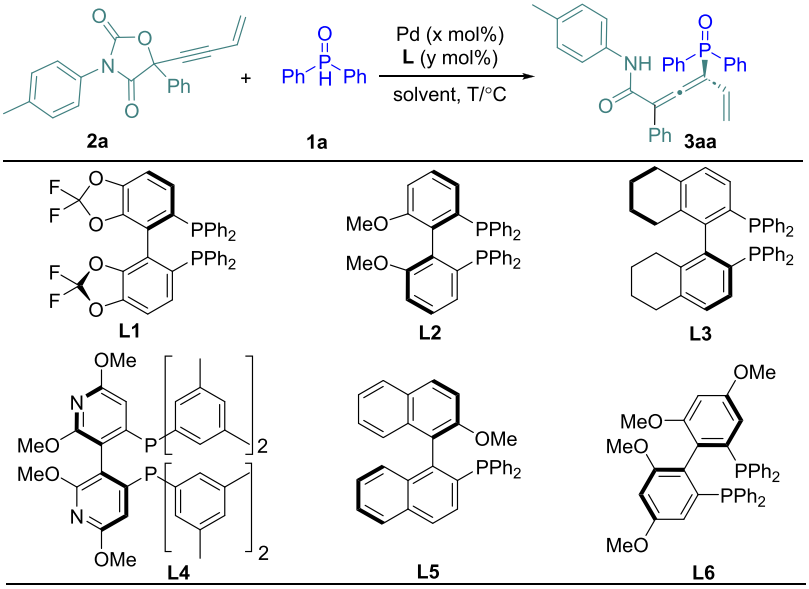


Figure 1. Palladium-catalyzed asymmetric allenyl C−P bond formations.

Under typical Pd-catalyzed reaction conditions, in the presence of secondary phosphine oxides (SPOs) and a propargylic precursor, the coordination of the [Pd/L*] complex preferentially occurs with the lone pair of the trivalent R^1R^2P−OH species rather than with the alkyne moiety. This is because the monodentate coordination of [Pd/L*] with the π−electrons of the alkyne is comparatively less stable (Figure 1c). To address this, the introduction of a terminal alkene allows for the formation of a more stable bidentate coordination with the [Pd/L*] complex. This suppresses the competitive coordination of R^1R^2P−OH, thereby enabling oxidative insertion into the propargylic substrate and leading to the formation of allenyl phosphine oxides.[12] This substrate design strategy is inspired by prior work from the Ma and Guo

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Table 1. Optimization of Asymmetric Allenylation Reaction[a]



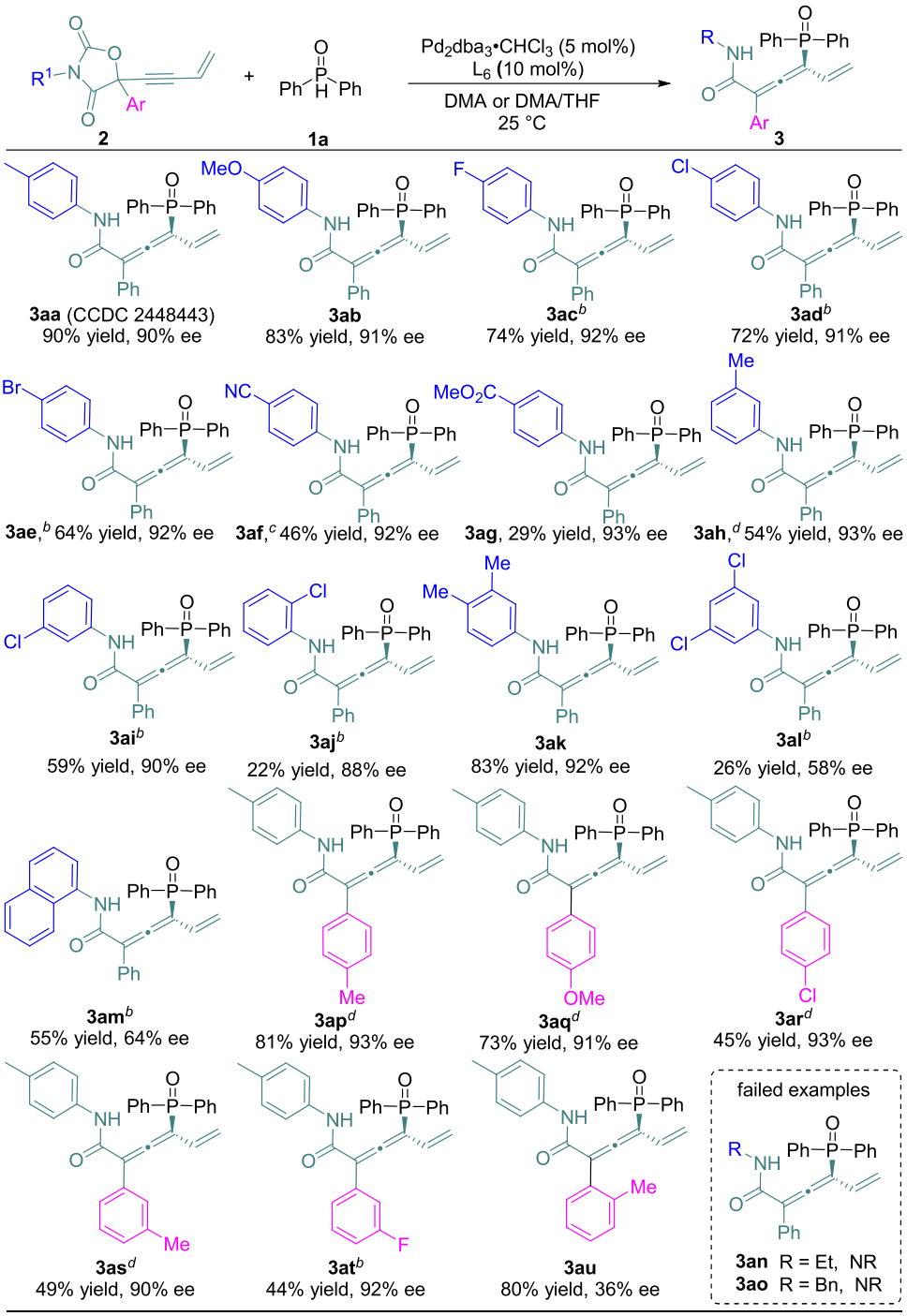
entry	L	solvent	T/°C	yield/%[b]	ee/%[c]
1	PPh_3	DCM	25	NR	
2	XantPhos	DCM	25	78	
3	DPEPhos	DCM	25	65	
4	DPPE	DCM	25	NR	
5	**L1**	DCM	25	84	31
6	**L2**	DCM	25	55	21
7	**L3**	DCM	25	55	−25
8	**L4**	DCM	25	20	−36
9	**L5**	DCM	25	6	21
10	**L6**	DCM	25	62	50
11	**L6**	THF	25	86	44
12	**L6**	EtOAc	25	98	27
13	**L6**	CH_3OH	25	66	74
14	**L6**	$CHCl_3$	25	13	40
15	**L6**	CH_3CN	25	47	78
16	**L6**	CH_3CN	0	15	90
17	**L6**	CH_3CN	−20	NR	
18	**L6**	DMF	25	39	92
19	**L6**	DMA	25	54	93
20	**L6**	DMSO	25	63	88
21[d]	**L6**	DMA	25	49	80
22	**L6**	DMA/EA = 1/1	25	75	91
23	**L6**	DMA/THF = 1/1	25	90	90

[a]Unless otherwise stated, all reactions were carried out with **1a** (0.1 mmol), **2a** (0.12 mmol), $Pd_2(dba)_3{\cdot}CHCl_3$ (5 mol %), and **L** (10 mol %) in 1 mL of solvent at 25 °C for 24 h. [b]Isolated yield. [c]Determined by chiral HPLC analysis. [d]$Pd_2(dba)_3{\cdot}CHCl_3$ (10 mol %) and **L6** (20 mol %) were used.

groups, who demonstrated that a similarly designed precursor could generate a vinylidene−π−allyl palladium(II) intermediate under Pd catalysis.[13] This intermediate was successfully intercepted by β-ketocarbonyl compounds to afford functionalized 1,2,3-butatriene derivatives (Figure 1d). As proposed, this unique π−allyl Pd intermediate can rearrange into a σ−π−σ form, rendering it susceptible to nucleophilic attack and enabling the synthesis of tetrasubstituted allenes. Motivated by our ongoing interest in the design and application of novel fungicide-inspired π−allyl Pd precursors particularly for the construction of tetrasubstituted axially chiral allenes,[14] we herein report a Pd-catalyzed enantioselective allenylation of SPOs. This transformation utilizes a newly designed, Vinclozolin-inspired enyne-type propargylic carbamate, providing new access to a class of highly functionalized, tetrasubstituted chiral allenes bearing allenyl amide, phosphine oxide, and alkene moieties (Figure 1e). This work differs fundamentally from pioneering Pd- or Ni-catalyzed systems in two key aspects: (i) The strategic incorporation of a terminal alkene group enables efficient generation of novel, reactive chiral vinylidene-π-allyl palladium intermediates for asymmetric construction of chiral allenes and (ii) the implementation of Vinclozolin-inspired enyne-type propargylic carbamates, featuring quaternary chiral centers, enabled direct construction of tetrasubstituted chiral allenes having phosphonyl unit.

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Scheme 1. Scope of 5-(But-3-en-1-yn-1-yl)-oxazolidine-2,4-diones 2[a]



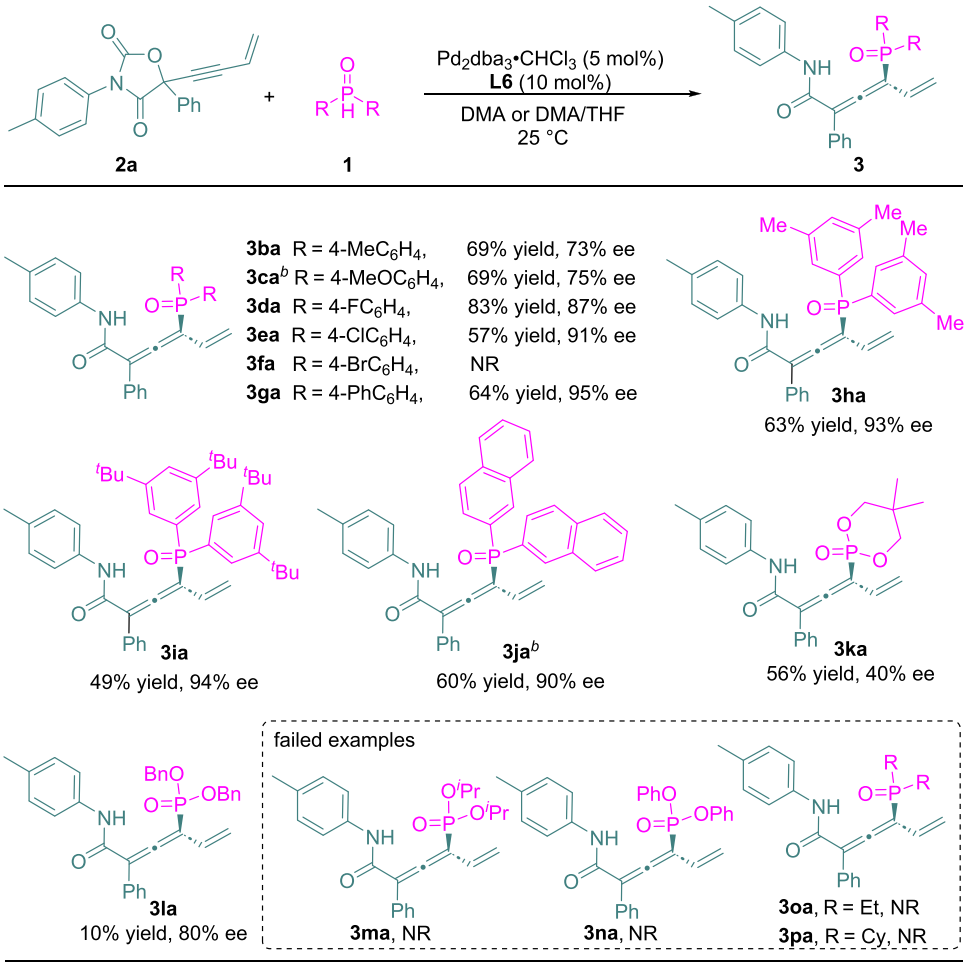
[b]Solvent: DMA/THF = 2/1. [c]Solvent: DMA/EA = 1/1. [d]Solvent: DMA. [a]Unless otherwise stated, all reactions were performed with **1a** (0.1 mmol), **2** (0.12 mmol), $Pd_2(dba)_3$•$CHCl_3$ (5 mol %), and **L6** (10 mol %) in 1 mL of solvent (DMA/THF = 1/1) at 25 °C for 37−65 h; all yields refer to isolated yield; ee was determined by chiral HPLC analysis.

RESULTS AND DISCUSSION

We initiated our study by selecting diphenylphosphine oxide **1a** and 5-(but-3-en-1-yn-1-yl)-5-phenyl-3-(*p*-tolyl)oxazolidine-2,4-dione **2a** as model substrates for the Pd-catalyzed allenylation reaction. Initial screening of racemic ligands (entries 1−4, Table 1) identified XantPhos and DPEPhos as effective, delivering the racemic product **3aa** in 78 and 65% yield, respectively. Subsequent evaluation of chiral ligands (entries 5−10 in Table 1 and **L7**−**L24** in Table S1) using $Pd_2(dba)_3$·$CHCl_3$ as the catalyst revealed that ligand **L6** afforded **3aa** in 62% yield with 50% ee (entry 10). Solvent screening (entries 11−15), including THF, EtOAc, MeOH, $CHCl_3$, and MeCN, identified MeCN as optimal, improving the ee to 78% albeit with a moderate yield of 47% (entry 15). Lowering the reaction temperature to 0 °C further enhanced the ee to 90%, though the yield decreased significantly to 15% (entry 16), while the reaction temperature of −20 °C completely suppressed reactivity (entry 17). Notably, explora-

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ACS Catal. 2025, 15, 17215−17224

Scheme 2. Scope of Secondary Phosphine Oxides 1[a]



[b]Solvent: DMA. [a]Unless otherwise stated, all reactions were carried out with **1a** (0.1 mmol), **2** (0.12 mmol), $Pd_2(dba)_3$•$CHCl_3$ (5 mol %), and **L6** (10 mol %) in 1 mL of solvent (DMA/THF = 1/1) at 25 °C for 40−48 h; yields refer to isolated yield; Ee were determined by chiral HPLC analysis.

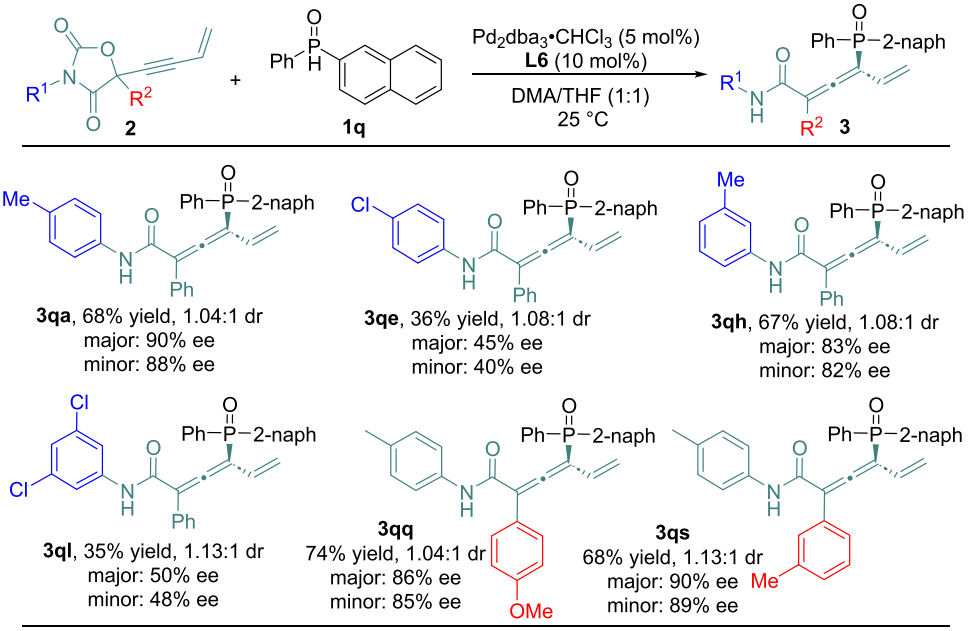
tion of polar solvents such as DMF, DMA, and DMSO (entries 18−20) indicated that DMA is particularly effective, giving **3aa** in 54% yield and 93% ee (entry 19). Increasing the catalyst and ligand loadings to 10 mol % of $Pd_2(dba)_3$·$CHCl_3$ and 20 mol % of **L6** yielded **3aa** in 49% yield with 80% ee (entry 21). Mixed solvent screening showed that the optimal solvent was a 1/1 mixture of DMA/THF, which furnished **3aa** in 90% yield and 90% ee (entry 23). The absolute configuration of **3aa** was unambiguously determined by a single-crystal X-ray diffraction analysis.

With optimized conditions established, a range of functionalized substrates **2** were evaluated (Scheme 1). Initial investigation of nitrogen substituents (R^1) revealed that substrates bearing either electron-donating (Me, OMe) or electron-withdrawing groups (F, Cl, Br, CN, and COOMe) at the *meta*- or *para*-position afforded the corresponding products in moderate to high yields (**3aa**−**3ai**, 29−90%) with consistently high enantioselectivity (90−93% ee). It was observed that para-substituted electron-deficient substrates tended to exhibit lower reaction efficiency, suggesting that the electronic properties of the amide moiety play a key role in the transformation. As illustrated in the proposed mechanism, this may be attributed to the amide functioning as a base to facilitate deprotonation of the SPO substrate. Sterically demanding substrates, *ortho*-chlorophenyl (**2j**) and 1-naphthyl-substituted substrate (**2m**), produced product **3aj** (22% yield, 88% ee) and **3am** (55% yield, 64% ee) in significantly reduced yield or ee, respectively. These results indicated that steric hindrance in the substrate exerted a pronounced negative effect on the efficiency and stereocontrol of the reaction. 3,4-Dimethylphenyl substrate **2k** performed well to provide product **3ak** in 83% yield with 92% ee, contrasting sharply with the performance of its dichlorophenyl analogue **2l** (its product, **3al**, 26% yield, 58% ee). Notably, *N*-alkyl-substituted variants such as *N*-ethyl (**2n**) and *N*-benzyl (**2o**) substrates proved to be unreactive. Variations of Ar groups were also investigated. Similar to the situation of R^1, *para*-substituted aromatics bearing electron-donating groups presented higher reactivity (**3ap**, **3aq**), compared with that of the electron-withdrawing group (**3ar**). *meta*-Methylphenyl and *meta*-fluorophenyl substrates were also compatible with the reaction to deliver the corresponding products **3as** and **3at** in both moderate yield and high ee. However, the *ortho*-tolyl-substituted substrate **2u** exhibited good reactivity but compromised enantioselectivity, likely due to steric constraints imposed by the *ortho*-methyl group.

Subsequently, we investigated the scope of secondary phosphine oxide **1** with respect to **2a**. As shown in Scheme

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ACS Catal. 2025, 15, 17215−17224

Scheme 3. Scope of 5-(But-3-en-1-yn-1-yl)-oxazolidine-2,4-diones 2 with respect to 2-Naphthyl Phenyl Phosphine Oxide 1q[a]



[a]Unless otherwise stated, all reactions were carried out with **1a** (0.1 mmol), **2** (0.12 mmol), $Pd_2(dba)_3{\cdot}CHCl_3$ (5 mol %), and **L6** (10 mol %) in 1 mL of solvent (DMA/THF = 1/1) at 25 °C for 21−48 h; all yields refer to isolated yield; dr was determined by ^1H NMR analysis; ee was determined by chiral HPLC analysis.

2, *para*-substituted diarylphosphine oxides bearing electron-donating (Me, OMe) or -withdrawing (F, Cl) groups all delivered products (**3ba**−**3ea**) in moderate to high yields (57−83%) with good enantioselectivity (73−91% ee). Unfortunately, the *para*-bromophenyl-substituted substrate (**1f**) did not give the desired product, possibly due to the competitive oxidative insertion of the Pd complex into aryl bromide. It is noteworthy that the *para*-biphenyl-substituted substrate **1g** displayed satisfactory reactivity and stereoselectivity in this reaction, affording product **3ga** in 64% yield with 95% ee. In addition, sterically encumbered substrates including bis(3,5-dimethylphenyl) (**1h**), bis(3,5-di-*tert*-butylphenyl)phosphine oxides (**1i**), and 2-naphthyl derivative **1j** proved viable, yielding products **3ha** (63%, 93% ee), **3ia** (49%, 94% ee), and **3ja** (60%, 90% ee) in good yields and excellent ee, respectively. Dialkylphosphine oxides, such as ethyl (**1o**) and cyclohexyl (**1p**), were found to be incompatible substrates. Phosphite substrates **1k**−**1n** were also tested in the current protocol, which exhibited limited compatibility: only diethyl (**1k**) and diisopropyl phosphites (**1l**) afforded **3ka** and **3la** with diminished yields and enantioselectivities.

Building upon successful axial chirality control, we investigated whether P-chirality could be simultaneously controlled by using diarylphosphine oxides with different substituents (Scheme 3). Under standard reaction conditions, 2-naphthyl phenyl phosphine oxide **1q** reacted with various alkenyl propargylic carbamates **2** to afford products **3qa**−**3qs** smoothly, but with generally poor diastereoselectivities. Among these, carbamate substrates bearing electron-donating groups displayed high to excellent enantioselectivity, whereas those with electron-withdrawing groups showed marginally affected ee values.

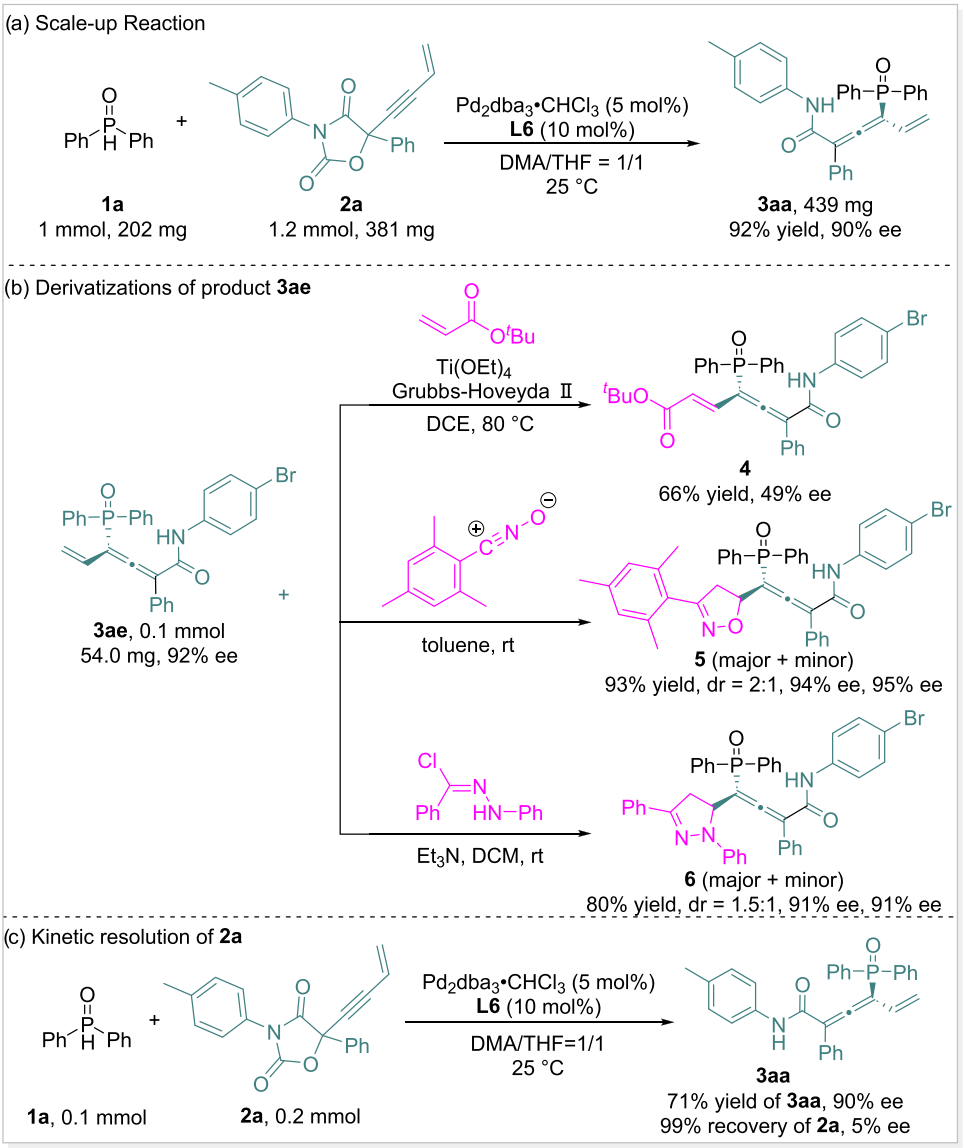
Furthermore, a scaled-up reaction of product **3aa** was performed to examine the utility of this protocol. As shown in Scheme 4a, the 1 mmol scale of the reaction proceeded smoothly, affording **3aa** in 92% yield with 90% ee. Both yield and enantioselectivity remained at high levels, demonstrating feasibility for large-scale production of chiral allenyl phosphine oxides. To establish the synthetic utility of the products, their further transformations were also explored. For instance, cross-metathesis of **3ae** with *tert*-butyl acrylate under catalysis of the Grubbs-Hoveyda II catalyst generated the *tert*-butoxyl ester-substituted product **4** in 66% yield with 49% ee (Scheme 4b). Subsequently, treatment of **3ae** (92% ee) with nitrile oxide or *N*-phenylbenzohydrazonoyl chloride both underwent [3 + 2] cycloadditions efficiently (Scheme 4b), giving product **5** (93% yield, 2:1 dr, 94% and 95% ee) and product **6** (80% yield, 1.5:1 dr, 91% and 91% ee), respectively. In a mechanistic investigation, a kinetic resolution experiment of **2a** produced **3aa** in 71% yield with 90% ee, and the recovered **2a** showed only 5% ee, revealing that no obvious kinetic resolution occurred in this process (Scheme 4c).

Subsequently, to probe the essence of the terminal alkene group in alkenyl propargylic carbamates, variants of substrate **2** were synthesized and evaluated in the current reaction. As shown in Scheme 5, under standard reaction conditions, propargylic carbamate **2v** provided only a trace amount of product as a racemate. The *n*-hexyl- and phenyl-substituted carbamates **2w** and **2x** showed unreactivity, and no desired allenylation products were detected. These results suggested that the terminal alkene group of carbamate **2** played a key role in this reaction.

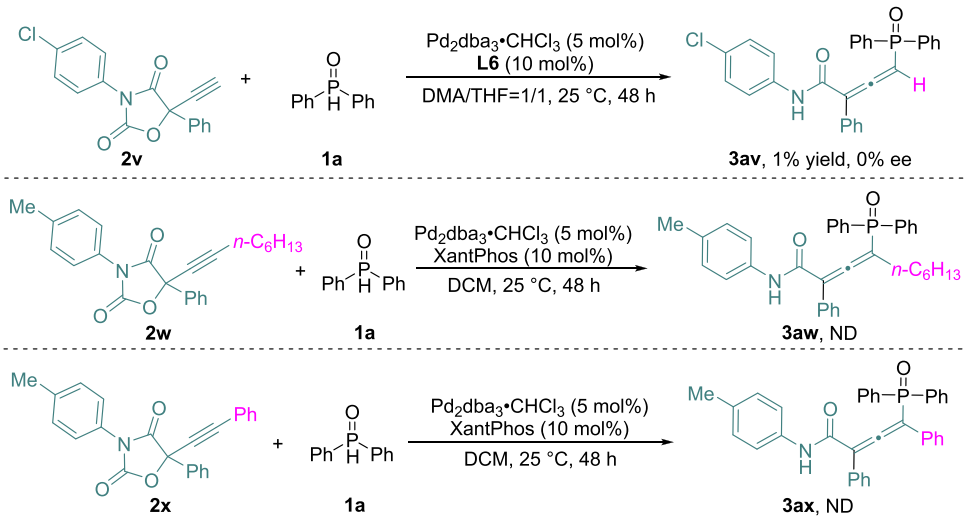
Based on the results of kinetic resolution experiments and substrate scope investigations of compound **2**, a plausible reaction mechanism for the Pd-catalyzed allenylation was proposed (Scheme 6). Initially, the Pd(0) precursor coordinates with ligand **L6** to form chiral Pd(0) complex **I**. This complex then engages the π-system of **2a** through a stable bidentate coordination involving both the alkyne and alkene moieties, affording intermediate **II**. Although the tautomeric

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ACS Catal. 2025, 15, 17215−17224

Scheme 4. Scale-Up Reaction, Derivatizations of Product 3ae, and Kinetic Resolution Study

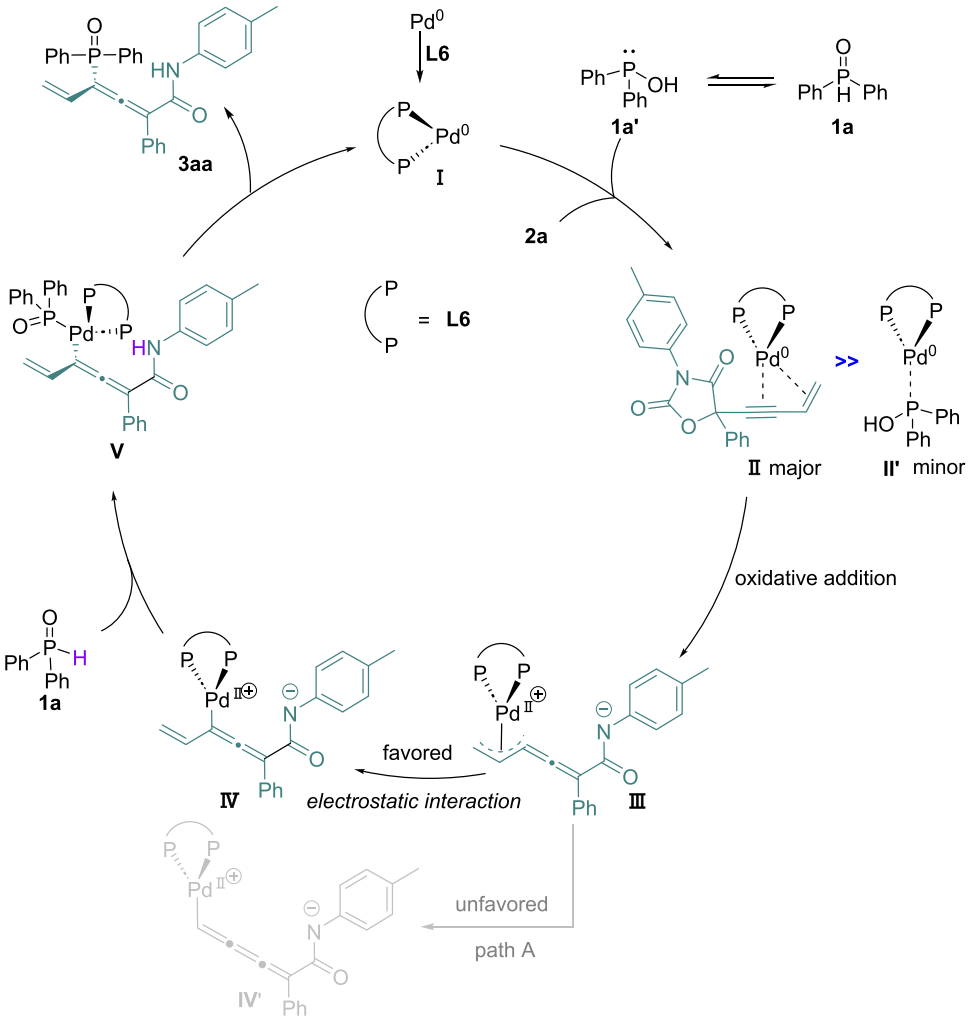


Scheme 5. Further Explorations on Variants of Substrate 2



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ACS Catal. 2025, 15, 17215−17224

Scheme 6. Plausible Reaction Mechanism



form of **1a** (**1a′**), a trivalent phosphorus species, could potentially coordinate with complex **I** to form an alternative intermediate **II′**, this pathway is considered minor due to the lower stability of **II′** relative to that of **II**. Subsequently, intermediate **II** undergoes oxidative addition accompanied by CO_2 extrusion, generating the key vinylidene−π−allyl palladium intermediate **III**. Electrostatic attraction between the Pd(II) cation and the anionic nitrogen center stabilizes intermediate **IV**, while the alternative conformation **IV′** is disfavored due to an unfavorable bond distance. Finally, deprotonation of **1a** by the basic nitrogen anion yields intermediate **V**, which undergoes reductive elimination to give product **3aa** and regenerate active chiral Pd(0) catalyst **I**.

CONCLUSIONS

In conclusion, we have developed a novel fungicide-inspired vinylidene−π−allyl palladium precursor that enables highly enantioselective allenylation of secondary phosphine oxides (SPOs), allowing efficient construction of axially chiral allenyl phosphine oxides in high yields with satisfactory enantioselectivities. Notably, the resulting chiral allene products feature two coordinating elements and a hydrogen-bonding site, highlighting their potential as versatile candidates for use as chiral catalysts and ligands. A key factor contributing to the success of this transformation is the enhanced bidentate coordination of the enyne-type propargylic carbamates with the chiral Pd complex. This interaction is sufficiently stable to outcompete the coordination of the trivalent tautomer of the SPOs, offering a promising strategy to overcome challenges associated with strongly coordinating substrates, such as pyridine-containing analogues in asymmetric Pd catalysis. Further mechanistic investigations and applications of this novel vinylidene−π−allyl palladium precursor are currently underway in our laboratory.

METHODS

General Procedure for the Preparation of Chiral Products 3. A flame-dried 10 mL Schlenk tube was charged with secondary phosphine oxides **1** (0.1 mmol, 1.0 equiv), enynes **2** (0.12 mmol, 1.2 equiv), $Pd_2dba_3{\cdot}CHCl_3$ (5 mol %, 5.2 mg), and **L6** (10 mol %, 6.4 mg), and then the vial was capped. After the mixture was evacuated and backfilled with argon three times, 1 mL of overdry solvent (see Schemes 1−5 for composition) was added. The reaction mixture was allowed to stir at 25 °C. Once the starting material was completely consumed (monitored by TLC), the reaction was quenched with H_2O. After the solvent was removed, the mixture was extracted with EtOAc. The combined organic layers were dried with Na_2SO_4, filtered, and concentrated in vacuo. The crude

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ACS Catal. 2025, 15, 17215−17224

material was purified by flash chromatography to afford products **3**.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c05571.

Experimental procedures, NMR spectra, HPLC chromograms, and crystal data for substrates and products (PDF)

Crystallographic data (CIF)

Accession Codes

CCDC 2448443 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: + 44 1223 336033.

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Notes

The authors declare no competing financial interest.

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